WORKED SOLUTION

STEP III 2022, Question 3

Pure • Implicit Differentiation

STEP III 2022, Question 3

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2022 STEP papers (a zip file):

www.ocr.org.uk/Images/696587-2022-past-papers.zip

The key idea

Differentiate implicitly. A stationary point needs the numerator of $\frac{\mathrm{d}y}{\mathrm{d}x}$ to vanish, which gives y in terms of x (or a linear relation between them). Substituting back into the curve's equation leaves a single equation in x, and the question becomes: when does that equation have two real roots? In (i) it is a quadratic in disguise; in (ii) it is a quartic, and the way to count its roots is to find its turning points.

Solution

Part (i)

Differentiating $ax^2 + bxy + cy^2 = 1$ implicitly with respect to x,

$$2ax + by + (bx + 2cy)\frac{\mathrm{d}y}{\mathrm{d}x} = 0.$$

At a stationary point $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$, so $2ax + by = 0$, that is $y = -\frac{2a}{b}x$ (we may divide by b because $abc \neq 0$). Substituting into the curve,

$$ax^2 - 2ax^2 + \frac{4a^2c}{b^2}x^2 = 1$$
$$\frac{a(4ac - b^2)}{b^2}x^2 = 1.$$

If $4ac - b^2 \leq 0$, the left-hand side is ≤ 0 for every x (recall $a > 0$), so there are no stationary points at all. So if the curve has two stationary points we must have $4ac - b^2 > 0$, that is $b^2 < 4ac$, as required. (Conversely, when $b^2 < 4ac$ we get $x = \pm|b|/\sqrt{a(4ac - b^2)}$, giving exactly two points.)

Ask yourself At these points, could $bx + 2cy$ also be zero, so that the derivative is not really defined?

If $2ax + by = 0$ and $bx + 2cy = 0$ together, then, since the determinant $4ac - b^2$ is non-zero, the only solution is $x = y = 0$, which is not on the curve. So the two points really do have $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$.

Part (ii)

Differentiating $ay^3 + bx^2y + cx = 1$ implicitly,

$$(3ay^2 + bx^2)\frac{\mathrm{d}y}{\mathrm{d}x} + 2bxy + c = 0.$$

At a stationary point, therefore, $2bxy + c = 0$. As $c \neq 0$, $x \neq 0$, and so $y = -\dfrac{c}{2bx}$. Substituting into the curve,

$$-\frac{ac^3}{8b^3x^3} - \frac{cx}{2} + cx = 1, \qquad \text{i.e.} \qquad -\frac{ac^3}{8b^3x^3} + \frac{cx}{2} = 1.$$

Multiplying through by $8b^3x^3$ and rearranging,

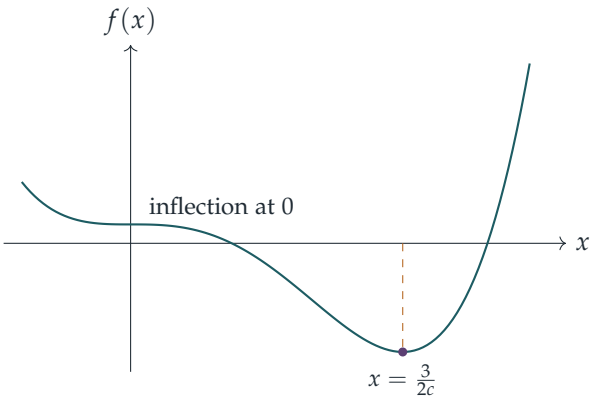
$$4cb^3x^4 - 8b^3x^3 - ac^3 = 0,$$

which is the required equation.

Call the left-hand side $f(x)$. Each stationary point has a different x-coordinate, because $y = -c/(2bx)$ is fixed by x. So two stationary points mean that f has at least two distinct real roots. To count the roots of f, look at its derivative:

$$f'(x) = 16cb^3x^3 - 24b^3x^2 = 8b^3x^2(2cx - 3).$$

The factor x^2 never changes sign, so f' changes sign only at $x = \frac{3}{2c}$. Suppose $c > 0$. Then the leading coefficient $4cb^3$ is positive (as $b > 0$), f is strictly decreasing for $x < \frac{3}{2c}$ and strictly increasing for $x > \frac{3}{2c}$; the point $x = 0$ is only a stationary point of inflection. Hence f has at most one root on each side of $\frac{3}{2c}$, and it has two distinct roots if and only if its minimum value is negative. If $c < 0$, everything is reversed: f increases then decreases, and has two distinct roots if and only if its maximum value is positive. In both cases, the condition is that $c\,f\left(\frac{3}{2c}\right) < 0$.



The case $a = -1$, $b = c = 1$: $f(x) = 4x^4 - 8x^3 + 1$ has a single turning point, a minimum at $x = \frac{3}{2}$, and two real roots.

Now evaluate at the turning point:

$$\begin{aligned} f\left(\tfrac{3}{2c}\right) &= 4cb^3 \cdot \frac{81}{16c^4} - 8b^3 \cdot \frac{27}{8c^3} - ac^3 \\ &= \frac{81b^3}{4c^3} - \frac{108b^3}{4c^3} - ac^3 = -\frac{27b^3 + 4ac^6}{4c^3}. \end{aligned}$$

So

$$c\, f\left(\tfrac{3}{2c}\right) = -\frac{4ac^6 + 27b^3}{4c^2},$$

and this is negative exactly when $4ac^6 + 27b^3 > 0$. Hence, if the curve has two stationary points, $4ac^6 + 27b^3 > 0$.

Part (iii)

The second and third equations say that both $2bxy + c$ and the coefficient $3ay^2 + bx^2$ of $\frac{\mathrm{d}y}{\mathrm{d}x}$ vanish, so this is a point of the curve where the method of (ii) breaks down. Since $c \neq 0$, the second equation gives $x \neq 0$ and $y = -\dfrac{c}{2bx}$. The first two equations are exactly what we used in (ii), so x satisfies

$$4cb^3x^4 - 8b^3x^3 - ac^3 = 0. \qquad (*)$$

Substituting $y = -c/(2bx)$ into the third equation,

$$\frac{3ac^2}{4b^2x^2} + bx^2 = 0 \quad \Longrightarrow \quad 4b^3x^4 = -3ac^2.$$

Multiplying by c, the first term of $(*)$ is $-3ac^3$, so $(*)$ becomes

$$-3ac^3 - 8b^3x^3 - ac^3 = 0 \quad \Longrightarrow \quad x^3 = -\frac{ac^3}{2b^3}.$$

Dividing $x^4 = -\dfrac{3ac^2}{4b^3}$ by this (allowed, as $x \neq 0$) gives

$$x = \frac{-3ac^2/(4b^3)}{-ac^3/(2b^3)} = \frac{3}{2c}.$$

So the solution sits exactly at the turning point of f from (ii). Putting $x = \frac{3}{2c}$ back into $x^3 = -\frac{ac^3}{2b^3}$:

$$\frac{27}{8c^3} = -\frac{ac^3}{2b^3} \quad \Longrightarrow \quad 54b^3 = -8ac^6 \quad \Longrightarrow \quad 4ac^6 + 27b^3 = 0,$$

as required. (Equivalently, $f\left(\frac{3}{2c}\right) = 0$ by the formula in (ii).) This is the boundary case of (ii): the two roots of the quartic have merged into a repeated root at the turning point, and the curve has a singular point there instead of two stationary points.

Checks

- In (i), take $a = c = 1$, $b = 1$: the curve $x^2 + xy + y^2 = 1$ is an ellipse, and $b^2 = 1 < 4$; its stationary points are at $x = \pm\frac{1}{\sqrt{3}}$, $y = \mp\frac{2}{\sqrt{3}}$, which do lie on it.
- In (ii), with $a = -1$, $b = c = 1$ we get $4ac^6 + 27b^3 = 23 > 0$ and f has minimum value $f(\frac{3}{2}) = -5.75 = -\frac{23}{4}$, matching $-\frac{27b^3+4ac^6}{4c^3}$.
- In (iii), choose $a = -2$, $c = 1.3$ and b with $27b^3 = -4ac^6$; then $x = \frac{3}{2c}$, $y = -\frac{c}{2bx}$ satisfies all three equations to rounding error. A computer check over thousands of random (a, b, c) also confirmed that the number of stationary points matches the conditions in (i) and (ii).

Common slips

- Forgetting that $x = 0$ is a critical point of the quartic: it is a point of inflection, not a turning point, and saying so is what lets you conclude there are at most two roots.
- Dropping the sign of c: the turning point is a minimum only when $c > 0$. Writing the condition as $c\,f(\frac{3}{2c}) < 0$ handles both cases at once.
- In (i), not noting that the case $b^2 = 4ac$ gives $0 = 1$, so it also has no stationary points.

Beyond STEP

The quantity $4ac^6 + 27b^3$ is, up to a factor, the *discriminant* of the quartic: it vanishes exactly when the quartic has a repeated root, just as $b^2 - 4ac$ does for a quadratic (and $4p^3 + 27q^2$ for the cubic $t^3 + pt + q$). Part (iii) shows the geometric meaning: a repeated root corresponds to a *singular point* of the curve, where both partial derivatives of $ay^3 + bx^2y + cx$ vanish together. Singular points and discriminants are the starting point of algebraic geometry; for a gentle, visual introduction to the related ideas about polynomials and curves, try Needham, *Visual Complex Analysis*, and for more problems in this style, Siklos, *Advanced Problems in Mathematics*.

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